[redacted]

VBF-T7R3-BOM-01 Case t7_r3 | 2026-08-26

Prepared: _____________ Reviewed: _____________ Approved: _____________

Virtual design package - simulation caliber (no physical build)

VBF-T7R3-BOM-01 Case t7_r3 | 2026-08-26 - BOM

Component	Mass g/cell	kg/kWh	Volume-fraction / basis	Source
Positive active material (NMC811)	16.8421620276	0.7488406291152013	active vol frac 0.665, layer 75.6 um	r7_v19_final_energy.json layer_kg_m2 x area (con
Positive conductive additive	Not modeled	Not modeled	no parameter in Chen2020; industry rec. ~2 wt%	parameter set gap (annotated)
Positive binder (PVDF)	Not modeled	Not modeled	no parameter; industry rec. ~2 wt%	parameter set gap (annotated)
Negative active material (graphite)	14.03934675	0.6242211205074475	active vol frac 0.75, layer 110 um	r7_v19_final_energy.json layer_kg_m2
Negative conductive additive	Not modeled	Not modeled	no parameter; industry rec. ~1 wt%	parameter set gap (annotated)
Negative binder (CMC/SBR)	Not modeled	Not modeled	no parameter; industry rec. ~3 wt%	parameter set gap (annotated)
Separator	0.259309284	0.011529477453533351	12 um x 0.53 (1-porosity 0.47)	r7_v19_final_energy.json layer_kg_m2
Electrolyte	7.21	0.3205729126149474	pore volume 6.004 mL x 1.2 g/cm3 (literature density), fill 1.0	derived (porosity x geometry); density = literature
Positive current collector (Al)	4.436640000000001	0.1972630522918142	16 um x 2700 kg/m3	Chen2020 set
Negative current collector (Cu)	11.042304	0.4909658190374043	12 um x 8960 kg/m3	Chen2020 set
Enclosure / tabs	Not modeled	Not modeled		parameter set gap (annotated)
TOTAL (contract, electrolyte excluded)	46.6197620616	2.0728200984054	mass_kg 0.0466198 x1000	energy.json:mass_kg
TOTAL incl. electrolyte	53.8297620616	2.393393011020348		derived (electrolyte excluded from contract)]
Cell energy (Wh)	22.49098322496203		integral V x I_1C dt, I_1C = 6.28 A	energy.json:energy_wh